Lecture 6

Last time: Requirements for vector spaces:

- $v+w$ and cv stay in the space
- linear combinations $cv+dw$ stay in the space.

Today: Column space : $Ax=b$
Null space

Column space of A is a subspace of $\mathbb{R}^4$.

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

$C(A)$ = all linear combinations of the columns.

Is $C(A)$ all of $\mathbb{R}^4$? NO!
Does $Ax=b$ have a solution for all b? NO!

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

4 equations
3 variables.

Which b's allow the system to be solved?

Can solve $Ax=b$ exactly when b is in $C(A)$.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Can solve if $b=0$.
if b is one of the columns.

$C(A)$ is a 2-dim'l plane in $\mathbb{R}^4$.
(col 3 = col 1 + col 2)

Span of a set of vectors: the subspace formed by all their linear combinations

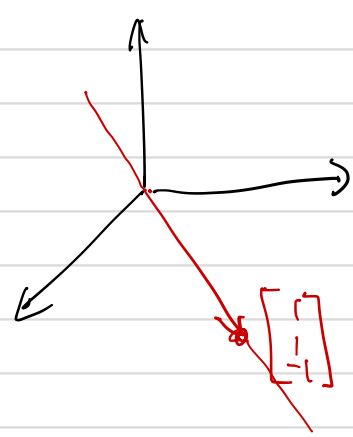
$$\left(\text{Span}(v_1, v_2, v_3) = C(B) \qquad B = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}. \right)$$

Null space of A is a subspace of $\mathbb{R}^3$.

$N(A)$ = all solutions $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ of $Ax=0$.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$N(A)$ contains $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} c \\ c \\ -c \end{bmatrix}$



Check that solutions to $Ax=0$ always give a subspace.

If $Av=0$ and $Aw=0$, then $\underbrace{A(v+w)}_{Av+Aw}=0$.

If $Av=0$, then $\underbrace{A(12\cdot v)}_{12\cdot Av}=0$.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

All solutions: $\underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{x_p} + c\underbrace{\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}}_{x_h}$.